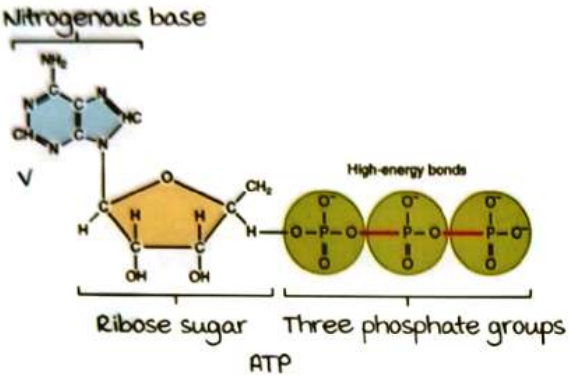
Www.Medicalstudyzone.com

----- Active space -----

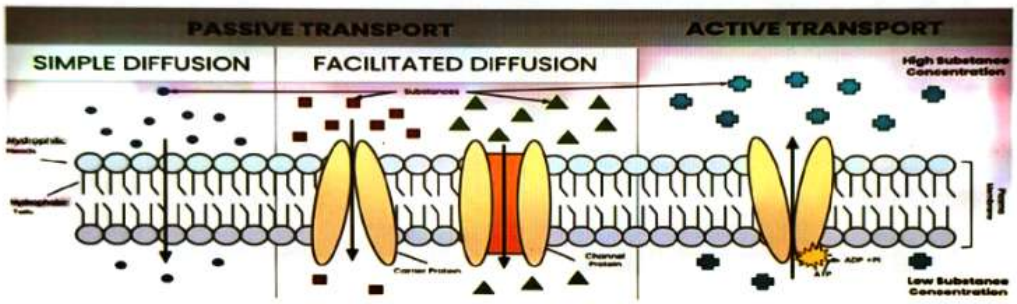
MEMBRANE TRANSPORT



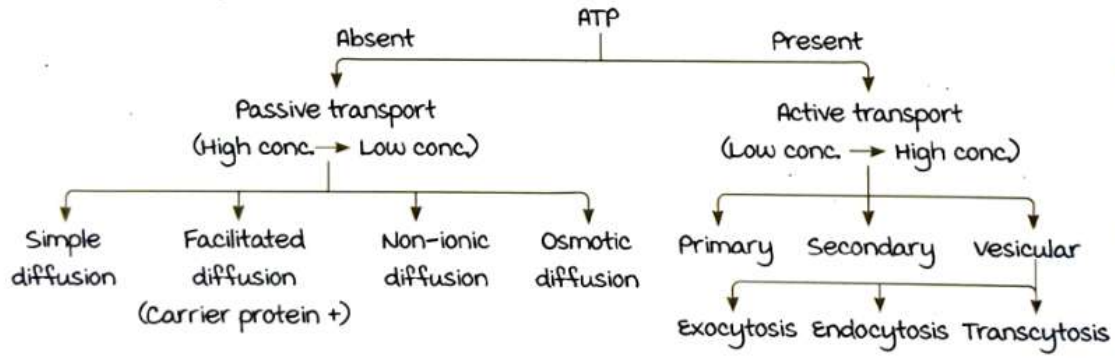
Awarded nobel prize for ATP discovery :

- Boyer
- Walker
- Jens skou

Classification :



Based on ATP requirement :

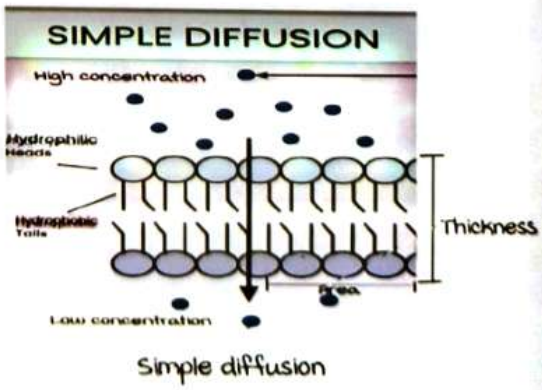


Passive Transport

00:04:33

SIMPLE DIFFUSION :

- Along concentration gradient.
- Without utilizing ATP.



Simple diffusion

Factors affecting transport :

membrane : Fick's law of simple diffusion.

Diffusion (D) α Concentration gradient (CG).

α Surface area (A).

α 1/Thickness (T).

Www.Medicalstudyzone.com

Substance :

----- Active space -----

Diffusion (D) α Solubility (Lipid soluble, Eg : O_2 diffusion across lung alveoli).

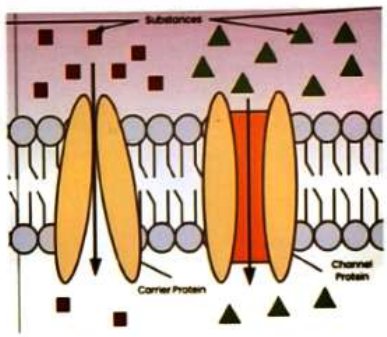
α 1/size.

Clinical application :

- Emphysema : ↓ surface area of membrane → ↓ diffusion of gases.
- Pulmonary fibrosis : Due to fibrosis → ↑ thickness of membrane → ↓ diffusion of gases.

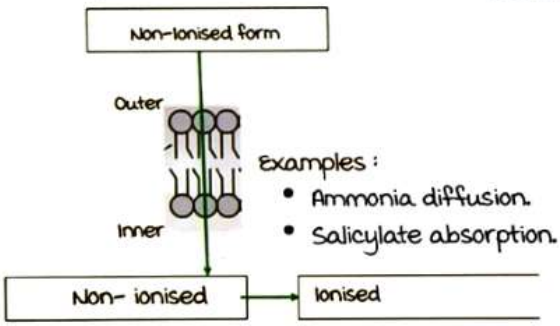
FACILITATED DIFFUSION :

- Through carrier protein :
  - Glucose transporter (GLUT) : Transport glucose.
  - Aquaporin : Transports water.
- Along concentration gradient.
- Without utilizing ATP.



Facilitated diffusion

NON-IONIC DIFFUSION :

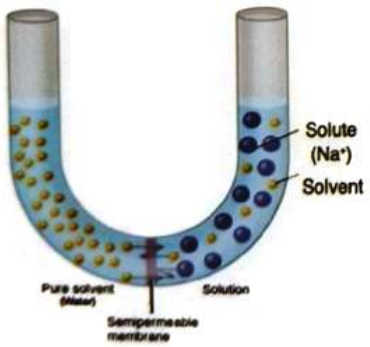


Examples :

- Ammonia diffusion.
- Salicylate absorption.

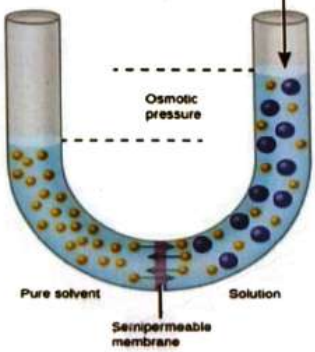
OSMOTIC DIFFUSION :

High H_2O → (Passive, Semipermeable membrane) → Low H_2O



Osmotic diffusion

Pressure applied to stop movement of H_2O



Osmotic pressure

Role of sodium :

Sodium (Osmotically active) : Drags H_2O.

Plasma osmolarity (Posm) :

Posm = 2 (Na^+) + 0.05 (Glu) + 0.36 (Blood urea nitrogen).

- Normal osmolarity of Na^+ : 300 mosm/L.

Feedback

Www.Medicalstudyzone.com

----- Active space -----

Note :

Osmoles : Concentration of osmotically active substance.

Osmolarity	Osmolality
mosm/L of solvent	mosm/Kg of solvent
most common	-

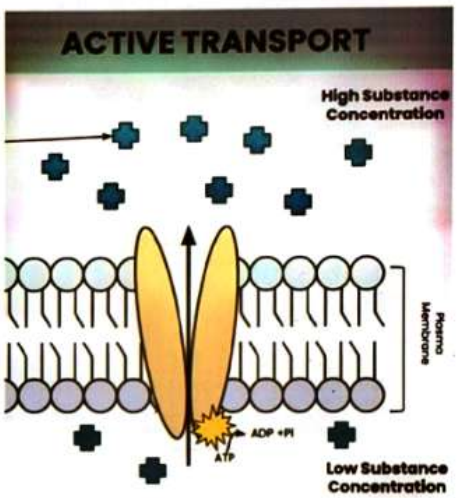
Active Transport

00:18:46

ATP utilized as source of energy.

Types :

1. Primary active transport.
2. Secondary active transport.



Active transport

PRIMARY ACTIVE TRANSPORT :

Transport channel have own ATPase.

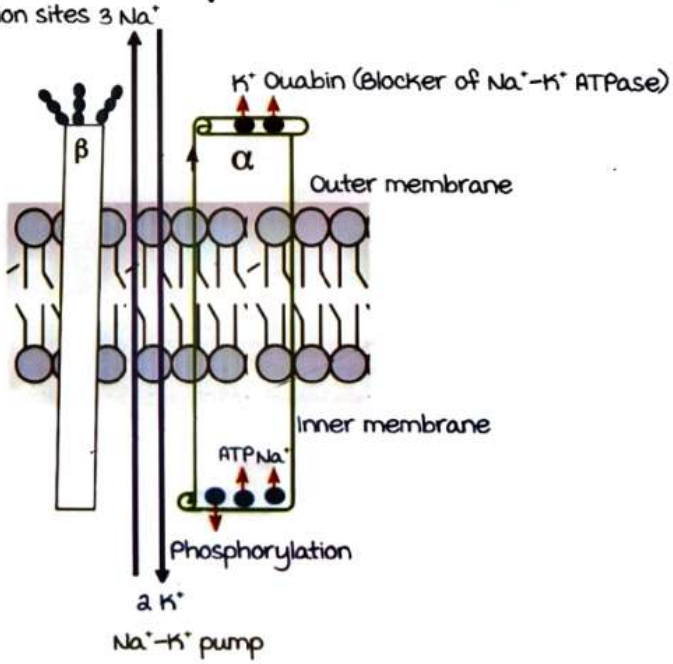
Examples :

- Na^+-K^+ ATPase.
- H^+-K^+ ATPase (Proton pump).
- Ca^{2+} ATPase.

Na^+-K^+ ATPase :

- Transmembrane protein in cell membrane.
- Heterogenous heterodimer (2 different subunits) : α and β.
- Coupling ratio : 3:2
  - movement of 3 Na^+ out and 2 K^+ inside cell : Electrogenic pump.

Structure : Glycosylation sites 3 Na^+



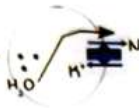
Feedback

Www.Medicalstudyzone.com

Functions :

----- Active space -----

1. Regulation of cell volume :

Cell swollen with H_2O →  → Cell shape return back to normal.

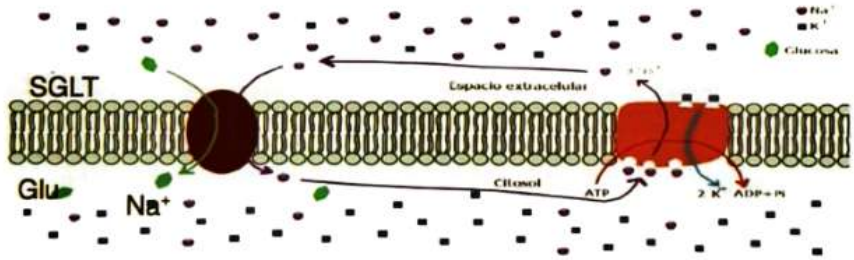
2. Resting membrane potential : Contribute 4mV (minor).
3. Indirect energy source : For secondary active transport.

Regulation : Activity of Na^+-K^+ pump.

	Increase [↑ ATP hydrolysis (metabolically active)]	Decrease
Examples	Thyroid hormone	Ouabin
	Insulin	Digitalis
	Aldosterone	Dopamine (Physiological)

SECONDARY ACTIVE TRANSPORT :

Secondary to energy gradient established by primary active transport.



Secondary active transport

Types :

1. Symporter : Both substances move in same direction.

Example		Site
SGLT (Sodium glucose linked transporter)	SGLT-1	GIT
	SGLT-2	PCT
Na^+-K^+-$2Cl^-$ cotransporter		Thick ascending limb of loop of henle
Na^+-I^- symporter		Thyroid gland

2. Antiport (Exchanger) : Substance move in opposite direction.

Eg : Cl^--HCO_3^- exchanger (Anion exchanger) in RBC.

VESICULAR TRANSPORT :

Active transport in the form of vesicles.
Irrespective of concentration gradient.
Help of Ca^{2+} and special protein.

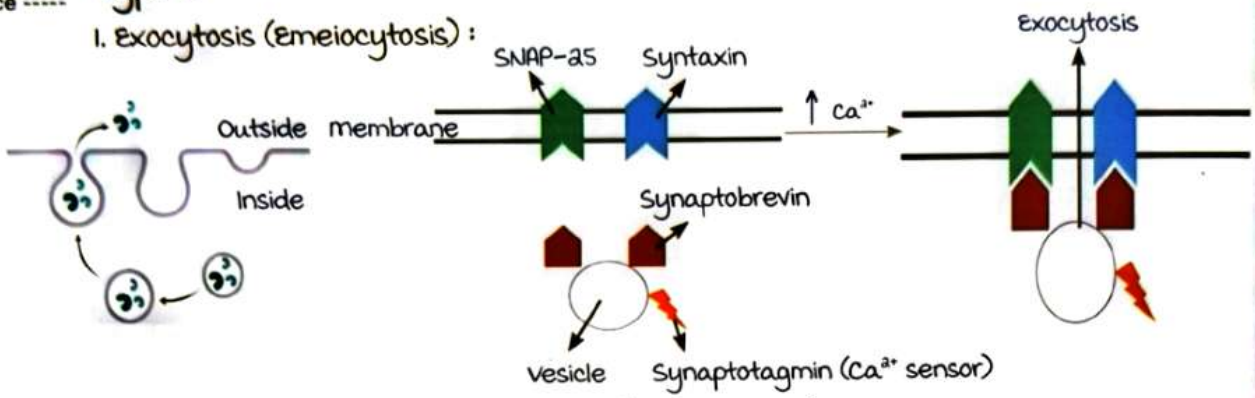
Feedback

Www.Medicalstudyzone.com

----- Active space -----

Types :

1. Exocytosis (Emeiocytosis) :

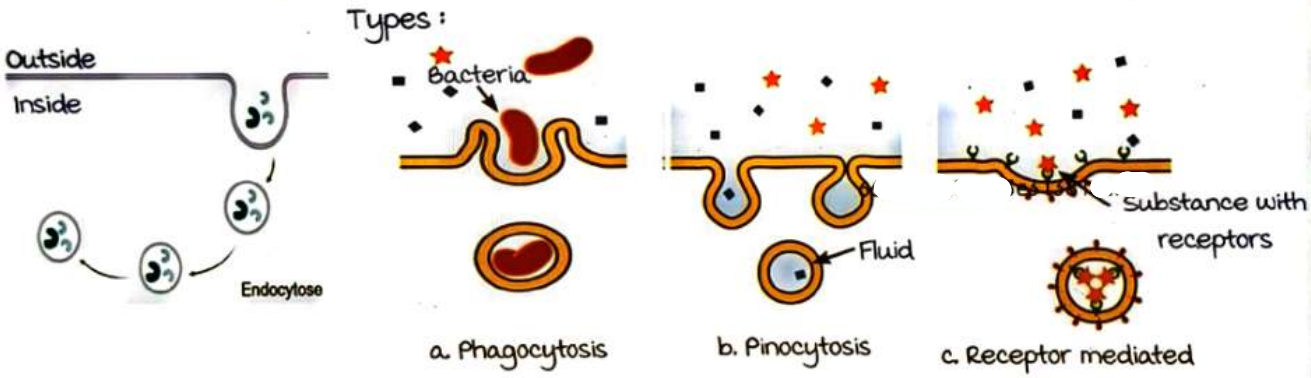


Snare proteins (Special proteins)

Note :

Botulinum toxin (To treat Achalasia cardia) → Breakdown synaptobrevin → (No exoxytosis) → Inhibit Ach release from vesicles → Flaccid paralysis

2. Endocytosis :



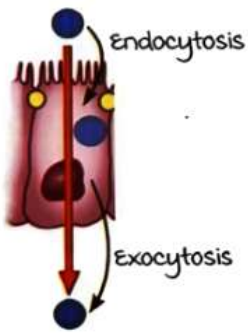
a. Phagocytosis

b. Pinocytosis

c. Receptor mediated endocytosis

Eg : LDL cholesterol with receptor proteins (Clathrin, Dynamin)

3. Transcytosis (Cytopempsis) :



Note :

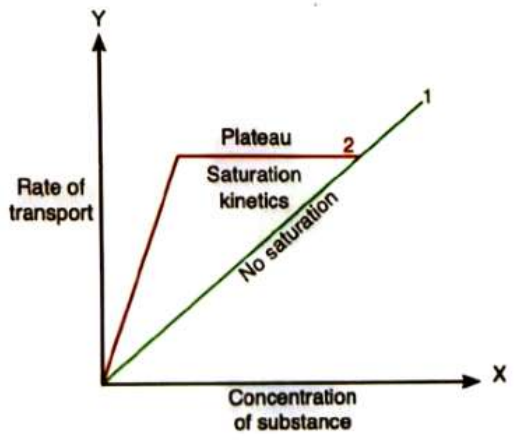
Saturation Kinetics :

Obeyed by :

- Active transport (Channel).
- Facilitated diffusion (Carrier protein).

Plateau occurs as synthesis of channel/ carrier protein takes time on ↑ demand.

- Not obeyed by : Simple diffusion.



Graph of saturation kinetics

Feedback